INCH-POUND
A-A-59423
21 May 1999

COMMERCIAL ITEM DESCRIPTION

VACUUM INTERFACE VALVE FOR SHIPBOARD USE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

1.1 Scope. This Commercial Item Description (CID) covers vacuum interface valves used with sewage vacuum, collection, holding, and transfer (VCHT) systems aboard ships. Vacuum interface valves control the discharge from urinals to the sewage vacuum collection piping. Although the requirements of this CID primarily refer to vacuum interface valves used with urinals, they may be applied to the acquisition of vacuum interface valves intended for use with other plumbing waste sources such as shower or galley drains.

2. SALIENT CHARACTERISTICS

2.1 Description. The vacuum interface valve (VIV) consists of an enclosure containing a discharge valve, liquid level sensing control assembly, vacuum check valve, interconnecting tubing, pipe, and couplings with hose clamps connecting the vacuum interface valve to the soil drain line.

2.2 Materials. The valve components shall be constructed of corrosion resisting materials.

2.3 Vacuum interface valve operation.

2.3.1 Automatic operation. Activation and operation of the VIV shall be fully automatic during normal operation. Flushing water from manually initiated flushometers or other sources flows through gravity drain piping to the VIV. The hydrostatic pressure of the rising flushing water level activates the VIV control mechanism which opens the discharge valve and allows the accumulated flushing water to be drawn into the sewage vacuum collection piping. The VIV controls activate and deactivate the flushing water discharge process. The controls shall operate pneumatically using only vacuum pressure available from the soil pipe connected to the sewage VCHT system. No electrical connections are permitted for activation or operation of the VIV. A check valve or other means shall be provided to prevent contaminating the VIV controls with sewage. The VIV discharge valve shall normally remain open for approximately 3 seconds. However, where the volume of flushing water is large, the vacuum interface valve shall cycle continuously until the flow is passed. Operation of the vacuum interface valve shall be smooth with no chattering due to discharged flushing water pressure and flow surges and operate without generating excessive noise.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Commander, Naval Sea Systems Command, SEA 03Q, 2531 Jefferson Davis Hwy., Arlington, VA 22242-5160.

AMSC N/A FSC 4510

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2.3.2 Manual operation. Means shall be provided to manually control the discharge valve to discharge the waste water to the sewage collection piping in the event of an inoperative vacuum interface valve control or inadequate vacuum pressure.

2.4 Discharge valve. The discharge valve shall be a pneumatically power-operated tubular diaphragm valve. The valve shall automatically close upon loss of system vacuum or failure of the valve operating mechanism. The tubular diaphragm shall be made of neoprene rubber.

2.5 Inlet and discharge connections. The vacuum interface valve waste inlet and discharge will be connected to shipbuilder supplied 1-1/2 inch nps soil pipe. The connection shall be made by a straight neoprene rubber coupling. Both ends of the coupling shall be securely clamped with stainless steel worm drive hose clamps. The connections shall not leak.

2.5.1 When specified (see 6.5), 90-degree elbow couplings shall be provided.

2.5.2 Unless otherwise specified (see 6.5), VIV inlet and discharge connections shall be as shown on figure 1.

2.6 Valve enclosure. The VIV enclosure shall be made of corrosion resistant material such as stainless steel or plastic as specified (see 6.5), and provided with a covered access for maintenance. Enclosure dimensions shall be as given on figure 1.

2.6.1 The VIV shall be installed foot mounted or supported by the adjoining piping as specified (see 6.5).

2.7 Performance.

2.7.1 Vacuum. The VIV shall operate connected to a soil or plumbing waste drain line that is under vacuum ranging from 14 to 18 inches of mercury (Hg).

2.7.2 Minimum vacuum. The VIV shall operate at a minimum vacuum of 9 inches Hg.

2.7.3 Flushing water. The flushing water shall be fresh water. When specified (see 6.5), flushing water shall be seawater. Where installed in galley drains the VIV shall pass galley food waste pulper discharge solids.

2.7.4 Endurance. The VIV shall operate by automatic operation for a minimum of 150,000 cycles with no failures or leaks.

2.7.5 Shock. When specified, (see 6.5), the VIV shall be shock tested and shall withstand the shock requirements for deck mounted equipment in accordance with MIL-S-901.

2.7.6 Airborne noise. When specified, (see 6.5),the VIV shall meet the grade B noise requirements of MIL-STD-470-1 during automatic operation.

2.8 Workmanship. Parts shall be of first class workmanship and shall be free from defects of manufacture which detract from their appearance or impair their serviceability.

2.9 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to

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metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request may be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3. REGULATORY REQUIREMENTS

3.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

4.2 Market acceptability (MA). The following market acceptability criteria shall be met to ensure the quality of the product to be provided under this CID. The offeror shall provide documentation with his bid that the product meets or exceeds all the market acceptability criteria. The contracting officer has the option of accepting or rejecting the product.

4.2.1 The company producing the item must have been producing a product meeting the requirements of this CID for not less than 2 years.

4.2.2 The company must have sold 500 units of the offered product, meeting this CID, in the commercial market place over the past 2 years. Where the product offered is a recently developed model and the number of units specified have not been sold, the offeror with his bid shall explain the differences between the previous model and the offered model and identify the advantages and improvements of the offered model.

4.2.3 The offeror shall provide supporting documentation as to the products serviceability and the long term (5 terms) availability of all necessary maintenance and logistics support parts and materials.

4.2.4 The offeror shall with his bid submit a list of ships and the number of units of the offered product installed, date (year) installed, and identify the installing activity.

4.4 Bid samples. When specified (see 6.5), one complete vacuum interface valve shall be submitted for testing by the Government. The bid sample shall be nonreturnable. Failure of any of the tests or examinations listed herein shall be cause for rejection of the bid.

4.5 Inspection requirements.

4.5.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5.2 through 4.5.10 of this document.

4.5.2 Failures. Failure of any of the tests and examinations listed herein shall be cause for rejection.

4.5.3 Examination. The vacuum interface valve shall be visually examined to determine compliance with requirements of this CID and with specified dimensions.

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4.5.4 Operational test. The VIV shall be connected to vacuum at a pressure specified herein and to flushing water. The VIV shall be operated not less than 10 times in the automatic discharge mode. All moving parts and mechanisms shall operate properly without any leaks or malfunction.

4.5.5 Minimum vacuum. The VIV shall be connected to vacuum at the minimum vacuum pressure specified herein, and to flushing water and operated not less than 10 times to demonstrate automatic operation with no failure to discharge.

4.5.6 Manual operation. The VIV shall be manually operated to demonstrate bypassing all controls to open the discharge valve and discharge the flush water.

4.5.7 Endurance cycling. The VIV offered by the bidder shall be connected to vacuum at system operating pressure and continuously cycled by automatic operation for the number of cycles specified in 2.7.4 with no failures or leakage. Minor leaks that can be corrected by tightening a connection may be repaired during the test. A failure of controls to operate or mechanism failure or leak shall be considered a failure and cause for rejection. Certification that this testing requirement has been met may be used in lieu of a test.

4.5.8 Shock test. The shock test when specified (see 6.5) shall be of the H.I. shock grade specified in the acquisition information. The shock test shall be performed with the VIV installed as required by the design. Foot mounted units shall be tested with the enclosure mounting feet bolted to a test fixture and inlet and discharge connections attached to pipe by contractor furnished straight couplings. Piping supported units shall be tested supported by the inlet and discharge connections attached to pipe by contractor-furnished straight couplings. During test the VIV inlet portion shall be filled with water.

4.5.8.1 Grade A. Grade A shock tested units shall exhibit no failure, permanent deformation, breakage, or any functional impairment. After shock test the unit shall be inspected and given the operational test specified herein.

4.5.8.2 Grade B. Grade B shock tested units shall not come loose from their mounts nor any parts become detached externally from the unit.

4.5.9 Airborne noise test. The noise test when specified (see 6.5) shall be performed with the VIV connected to vacuum at a pressure specified herein and to flushing water and operated in the automatic discharge mode to determine conformance with requirements specified in 2.7.6.

5. PACKAGING

5.1 Packaging. Preservation, packing, labeling, and marking shall be as specified in the contract or order.

6. NOTES

6.1 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

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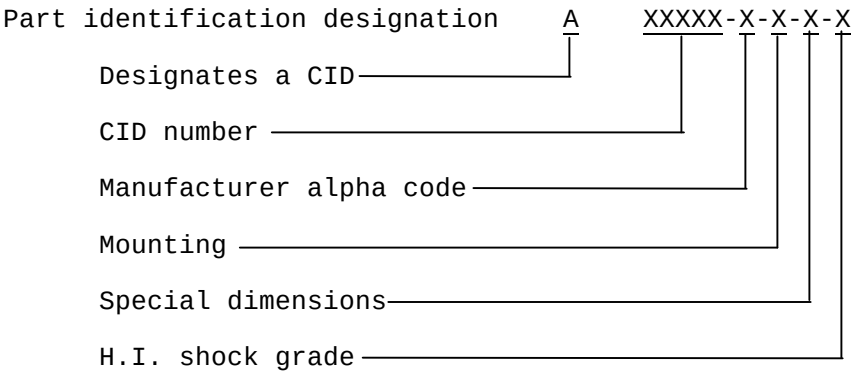

Manufacturer alpha code	Installation mounting	Special dimensions	H.I shock grade
A-	1- Foot mounted	N-No	A-grade A
B-	2- Pipe supported	Y-Yes	B-grade B
C-			C-Not tested
D-			
E-			

6.2 Source of documents.

6.2.1 Federal Acquisition Regulations (FAR) may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

6.2.2 Federal and Military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

6.3 Sources of supply. Manufacturers whose products are known to meet the requirements of this CID are listed below. Competition is not limited to those products and manufacturers.

Drawing number 7250004
Envirovac Inc.
1260 Turret Drive
Rockford, IL 61115-1486

Model AVV100
Jered Brown Brothers Inc.
1608 Newcastle Street
P.O. Box 904
Brunswick, GA 31521-0904

6.4 Intended use. Vacuum interface valves covered by this CID are intended for use in VCHT systems on new construction ships or as stock replacements for VIVs currently installed on existing ships. When acquiring VIVs for replacement purposes, the contracting activity must ensure that interface requirements are specified for inlet connections, outlet connections, and mounting feet (if used), which are compatible with the VIV being replaced to avoid the need for costly ship modifications. Acquisition for new construction allows for more latitude in VIV dimensions and interface requirements, enabling more suppliers to bid thereby increasing competition.

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6.5 Ordering data. Purchasers should specify the following:

(a) Title, number, and date of this commercial item description.
(b) If bid samples are required.
(c) If 90 degree couplings are required.
(d) If special connections are required.
(e) Enclosure material.
(f) If installation by foot mounting or pipe support is required and the bolting pattern.
(g) If seawater flushing is required.
(h) H.I. shock grade required.
(i) If airborne noise test is required.
(j) Levels of preservation, packing, labeling, and marking required.
(k) When drawings are required.
(1) When commercial manuals or basic manuals are required.

6.6 National stock numbers (NSNs). The following is a list of NSNs assigned which correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

NSN	ITEM
4820-01-284-4900	7250004

6.7 Bid information. The contracting officer should require that the offeror with his bid provide sufficient information to clearly depict the product offered. This information may be in the form of sketches, drawings, catalog sheets or other documents.

6.8 Drawing requirements. The requirement for drawings should be considered when this specification is applied on a contract. The data required shall be developed as specified by an approved Data Item Description (DID Form 1664) titled Commercial Drawings and Associated Lists, DID No. DI-DRPR-81003 and delivered in accordance with the approved CDRL incorporated into the contract. The data item description will be approved into the contract. The data item description will be approved and listed in DOD 5000.19L, Vol. II AMSDL.

6.10 Subject term (key word) listing.

Marine sanitation device (MSD)
Plumbing fittings
Pollution control system
Sewage system
Urinal
Vacuum collection, holding and transfer (VCHT)
VCHT piping system

Preparing activity:
Navy - SH
(Project 4510-0301)

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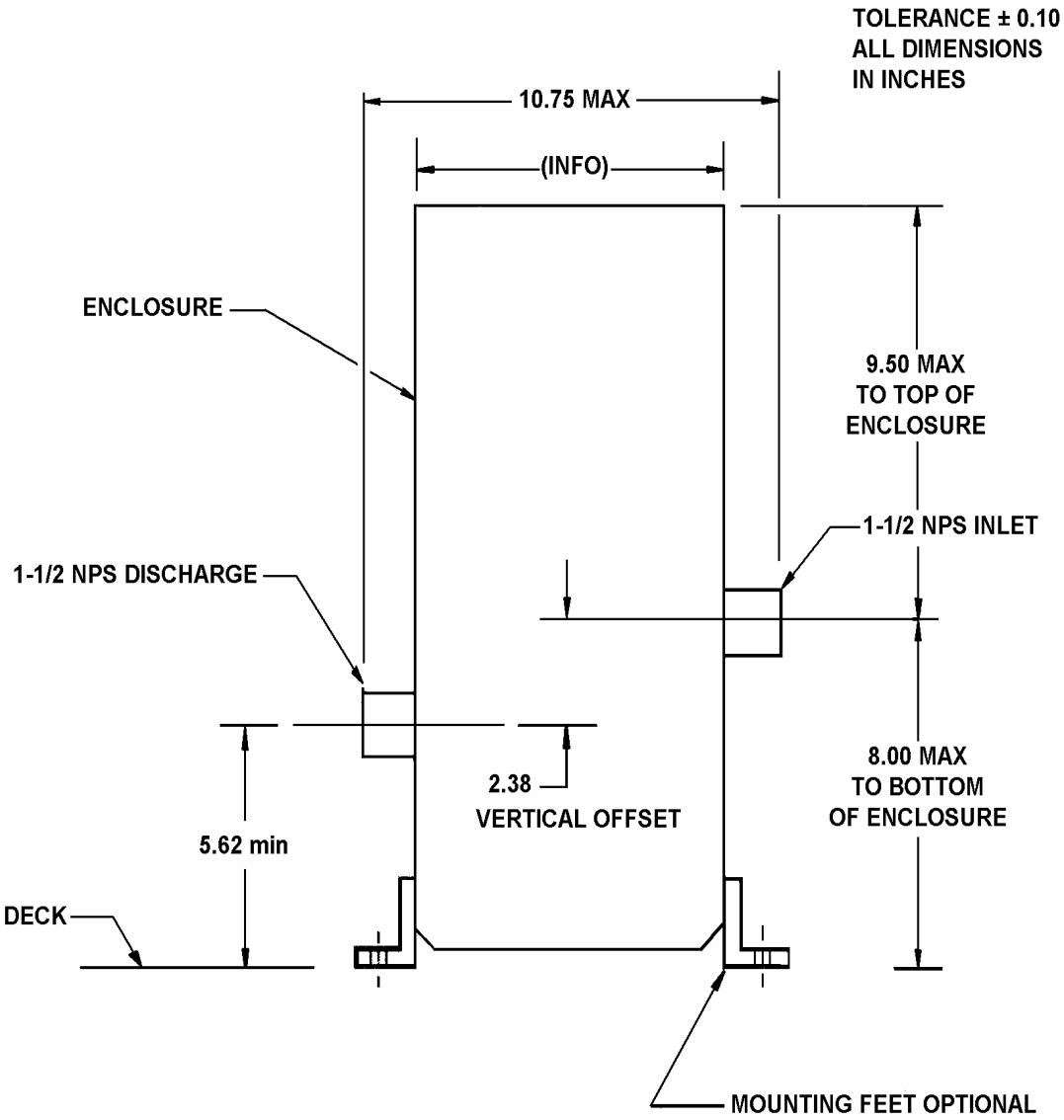

FIGURE 1. Vacuum interface valve installation outline.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4,5,6, and 7 and send to preparing activity.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-59423	2. DOCUMENT DATE *(YYYYMMDD)* 1999/05/21

3. DOCUMENT TITLE
VACUUM INTERFACE VALVE FOR SHIPBOARD USE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(if applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME John Julian, Code SEA 03L1	b. TELEPHONE *(Include Area Code)* (1) Commercial (703) 602-0351 X150 (2) DSN
c. ADDRESS *(Include Zip Code)* Commander, Naval Sea Systems Command ATTN: SEA 03Q, 2531 Jefferson Davis Hwy Arlington, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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